JAVA COLLECTIONS CASE STUDY 3

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Unique Student IDs - HashSet

Topic: Set Interface & HashSet Class
Environment: Java SE 17+ / 24

1. Problem Statement & Scenario Description

Scenario: A college wants to maintain unique student IDs. If the same ID is entered more than once, it should not be stored again. The system must register IDs, verify existing registration, and report total unique students.

#	Required Operation / Method	Implementation Details & Collections Logic
1	**registerStudent(Set ids, int id)**	Delegates to ids.add(id); returns true if newly added, or false if ID already registered.
2	**isRegistered(Set ids, int id)**	Performs fast membership query using ids.contains(id) returning boolean status.
3	**totalStudents(Set ids)**	Returns total count of distinct registered student IDs using ids.size().

2. Collections Framework Concepts & Technical Analysis

- **HashSet Uniqueness Guarantee:** Enforces mathematical set semantics; duplicate elements cannot exist in a HashSet.
- **O(1) Constant Time Complexity:** Offers constant-time performance for basic operations (add, contains, size) assuming proper hash dispersion.
- **Automatic Duplicate Rejection:** The add() method automatically detects collisions using hashCode() and equals() without throwing exceptions.

3. Complete Java Solution Code (Human Written - No Comments)

Clean, idiomatic Java implementation completed according to instructions with zero comments.

Source File: src/collections_case_studies/StudentRegistration.java

```
package collections_case_studies;

import java.util.HashSet;
import java.util.Set;

public class StudentRegistration {

    public static boolean registerStudent(Set<Integer> ids, int id) {
        return ids.add(id);
    }

    public static boolean isRegistered(Set<Integer> ids, int id) {
        return ids.contains(id);
    }

    public static int totalStudents(Set<Integer> ids) {
        return ids.size();
    }

    public static void main(String[] args) {
        Set<Integer> ids = new HashSet<>();

        System.out.println(registerStudent(ids, 101));
        System.out.println(registerStudent(ids, 102));
        System.out.println(registerStudent(ids, 103));
        System.out.println(registerStudent(ids, 101));

        System.out.println("Registered: " + isRegistered(ids, 102));
        System.out.println("Total Students: " + totalStudents(ids));
    }
}
```

4. Program Execution & Verification Output

Console output produced by running the completed driver code matching expected laboratory output:

Console Standard Output (stdout): java collections_case_studies.StudentRegistration

```
true
true
true
false
Registered: true
Total Students: 3
```